Sangbum Daniel Choi

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Summary

Machine-learning researcher and engineer specializing in multimodal pre-training, text-image alignment, post-training, model evaluation, and production systems. Led foundation-model projects and built distributed training and evaluation infrastructure across the model-development lifecycle.

Experience

- **Toss Bank** — Seoul, South Korea
  Data Scientist — *Jan 2026 - Present*
  - [**Lead**] **End-to-End Document AI**: Led document extraction from research through production with a 1-billion-parameter vision-language model for multi-page classification, orientation, layout detection, optical character recognition, and table understanding, paired with a 35-billion-parameter mixture-of-experts language model (3 billion active parameters) for key-value extraction. Post-trained the vision-language model, built stage-level diagnostics, and reached 61% exact-match accuracy.
  - **Language Model Evaluation**: Built a benchmark using public evaluation protocols and scoring criteria to compare Qwen3.5-397B, Qwen3.8, GLM-5.3, and DeepSeek V4 Flash 0731. Measured agentic capability and Korean-language understanding.
  - [**Lead**] **AI Authentication**: Developed an identity-card recognition and facial-verification model for real-time mobile inference. Validated on-device performance using Open Neural Network Exchange (ONNX) and WebGPU.
  - **Enterprise AI Answering Agent**: Built an internal Slack answering agent using Qwen3-0.6B and tool calling. Curated fine-tuning data from internal knowledge notes, Slack threads, team knowledge bases, and code and runbooks in 31 GitHub repositories, removing personally identifiable information. Tracked source region, topic, task type and complexity, token count, data origin, and whether answers came from ground truth or a teacher model; built segmented loss analysis to surface training issues. Added a demo bot that routes between hosted large language models and a fine-tuned model running locally on Mac; recorded thumbs-up and thumbs-down feedback to guide later fine-tuning.
  - **On-premise AI Coding Agent**: Designed planner/critic/executor orchestration for coding agents using GLM-5.3 and Qwen3.5-397B. A task router selects a small-model executor for simple tool calls, including Kubeflow jobs, or a large-model executor for complex coding. Three critics review results and vote by majority to resolve disagreements and decide when to loop for revisions. Added persistent-memory knowledge agents for internal domain questions the isolated environment could not answer through web search, plus specialist agents for translation, MLflow, and pull-request review; integrated OpenCode command-line and text interfaces with Kubeflow.
- **SuperbAI** — Seoul, South Korea
  Machine Learning Engineer — *Sep 2021 - Jan 2026*
  - [**Lead**] **Multimodal Foundation Model**: Led two machine-learning engineers through multimodal pre-training and staged text-image alignment for a visual-grounding model; curated 1.1 million images and led evaluation and deployment.
  - **Post-training and Efficiency**: Applied low-rank adaptation and adapter tuning with memory-efficient training, reducing graphics processing unit (GPU) memory use by 65.6% and training time by 44.3% while maintaining model performance.
  - [**Lead**] **Machine Learning Training and Serving Platform**: Built distributed graphics processing unit (GPU) training, evaluation, and deployment infrastructure with PyTorch and AWS Batch. Improved TensorRT and Triton serving throughput fivefold and delivered more than 1,100 models to over 60 customer endpoints in one year.
  - **Computer Vision and Pattern Recognition 2025 Challenges**: Placed second and fourth in two object-detection challenges; evaluated the model on public benchmarks.
  - **Interactive Segmentation**: Built browser, hybrid, and cloud-based segmentation tools using RepViT-SAM, FocalClick, and Segment Anything models, increasing labeling speed twenty-fivefold.
- **Kakao Enterprise** — Pangyo, South Korea
  Machine Learning Engineer Intern — *Feb 2021 - Aug 2021*
  - **Automated Machine Learning**: Fixed a scaling issue in AutoGluon NeuralNetFastAI and developed a Flask-based model-training and inference framework.

- **Team ISLAND** — Seoul, South Korea
  Chief Technology Officer — *Mar 2019 - Oct 2020*
  - **Startup**: Co-founded the company as Chief Technology Officer, raised $290,000, and managed a five-developer core team across Android, Unity, and deep learning; led teams of up to eight people.
  - **Mobile Machine Learning**: Built ZZAZZ, a mobile video editor using detection, segmentation, three-dimensional mapping, and tracking; deployed lightweight vision models through TensorFlow Lite.

Open Source

- **OpenLocalAgent**: Building tool-calling agents from scratch in PyTorch using open data, with pre-training, mid-training, post-training, and evaluation across ten public agent benchmarks.
- **Optical Character Recognition**: Building a reproducible evaluation and improvement framework for document vision-language models with fewer than one billion parameters, covering accuracy, calibration, robustness, and low-rank adaptation.
- **Absorbed**: Developing an open-source SwiftUI roguelike with turn-based combat, branching expeditions, spirit capture, persistent progression, and English/Korean localization.
- **Hugging Face Transformers**: Contributed more than 40 pull requests to the Transformers codebase across multimodal and vision models, shipping model architectures, processors, conversion scripts, distributed-training fixes, tests, and documentation.
- **Hugging Face Transformers - Segment Anything 2**: Led the integration of Segment Anything 2 into Transformers, including model support, image and video processors, conversion tools, documentation, and tests; credited in the official model documentation.
- **Hugging Face Transformers - Molmo2**: Opened Molmo2 support for Transformers and published danelcsb/Molmo2-4B on the Hugging Face Hub.

Papers

- **ZERO: Multi-modal Prompt-based Visual Grounding**: arXiv - Jul 6, 2025
- **MobileHumanPose: Toward real-time 3D human pose estimation in mobile devices**: IEEE/CVF CVPR Workshop - Jun 19, 2021
- **Improved DOA estimation of MUSIC algorithm based on irregular microphone array**: UIUC Research - Dec 20, 2018
- **On Joint Transfer of Energy and Information: A Markov Decision Problem Formulation**: Research paper - Dec 13, 2018
- **Terahertz Signal Analysis in Tissue of Gastroesophageal Reflux Disease**: Sejong Science High School - Dec 31, 2013

Education

- **KAIST (Korea Advanced Institute of Science and Technology)** — Daejeon, South Korea
  Professional Master in Electrical Engineering; Grade Point Average: 3.55/4.3 — *Feb. 2020 – Aug. 2021*
- **POSTECH (Pohang University of Science and Technology)** — Pohang, South Korea
  Bachelor in Electrical Engineering; Grade Point Average: 3.4/4.3 — *Mar. 2015 – Feb. 2020*
- **UIUC (University of Illinois Urbana-Champaign)** — Illinois, United States
  Exchange Student; Grade Point Average: 3.6/4.0 — *Jan. 2018 – Dec. 2018*

Technical Skills

- **Model Development**: Python, PyTorch, Transformers, multimodal pre-training, text-image alignment, post-training, low-rank adaptation, adapter tuning, evaluation
- **Systems**: Linux, AWS Batch, distributed model training, MLflow, TensorRT, Triton, ONNX, TensorFlow Lite, Git
- **Agents**: LangChain, Langfuse, Claude Code, Codex, OpenCode
- **Languages**: Korean (native), English (professional)